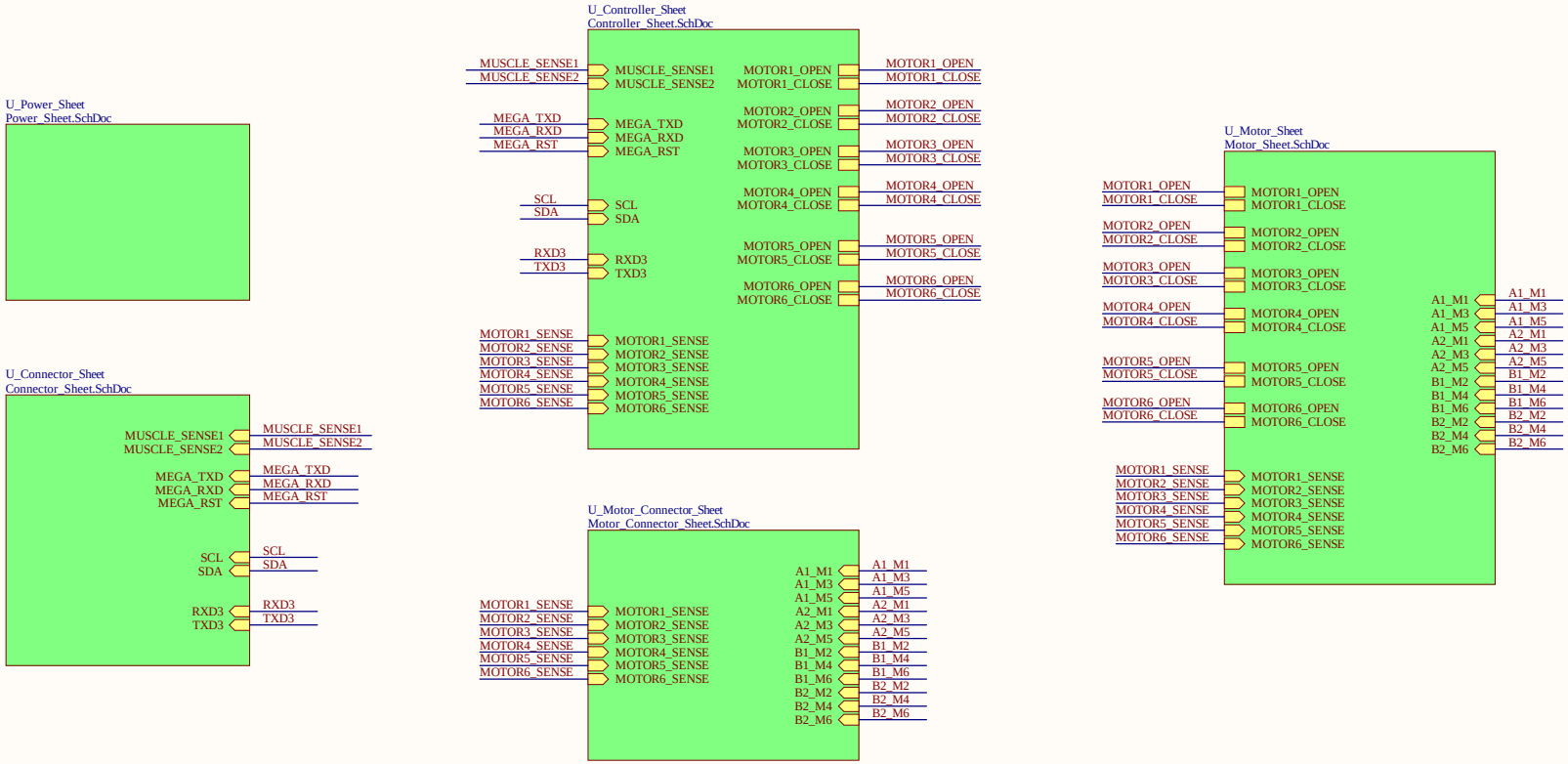
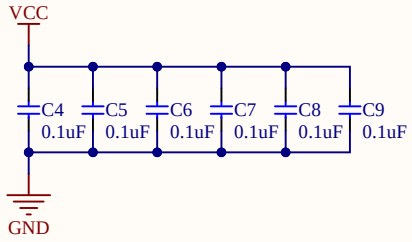
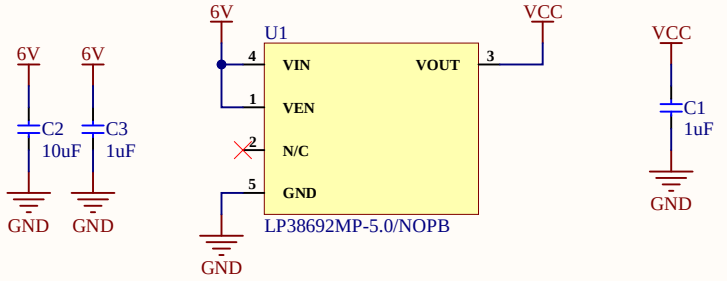

U_Power_Sheet
Power_Sheet.SchDoc
U_Connector_Sheet
Connector_Sheet.SchDoc
MUSCLE_SENSE1
MUSCLE_SENSE2
MEGA_TXD
MEGA_RXD
MEGA_RST
SCL
SDA
RXD3
TXD3
U_Controller_Sheet
Controller_Sheet.SchDoc
MOTOR1_OPEN
MOTOR1_CLOSE
MOTOR2_OPEN
MOTOR2_CLOSE
MOTOR3_OPEN
MOTOR3_CLOSE
MOTOR4_OPEN
MOTOR4_CLOSE
MOTOR5_OPEN
MOTOR5_CLOSE
MOTOR6_OPEN
MOTOR6_CLOSE
MOTOR1_SENSE
MOTOR2_SENSE
MOTOR3_SENSE
MOTOR4_SENSE
MOTOR5_SENSE
MOTOR6_SENSE
U_Motor_Connector_Sheet
Motor_Connector_Sheet.SchDoc
A1_M1
A1_M3
A1_M5
A2_M1
A2_M3
A2_M5
B1_M2
B1_M4
B1_M6
B2_M2
B2_M4
B2_M6
U_Motor_Sheet
Motor_Sheet.SchDoc

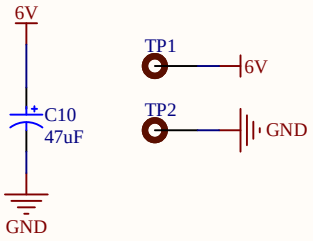
DECOUPLING CAPACITORS



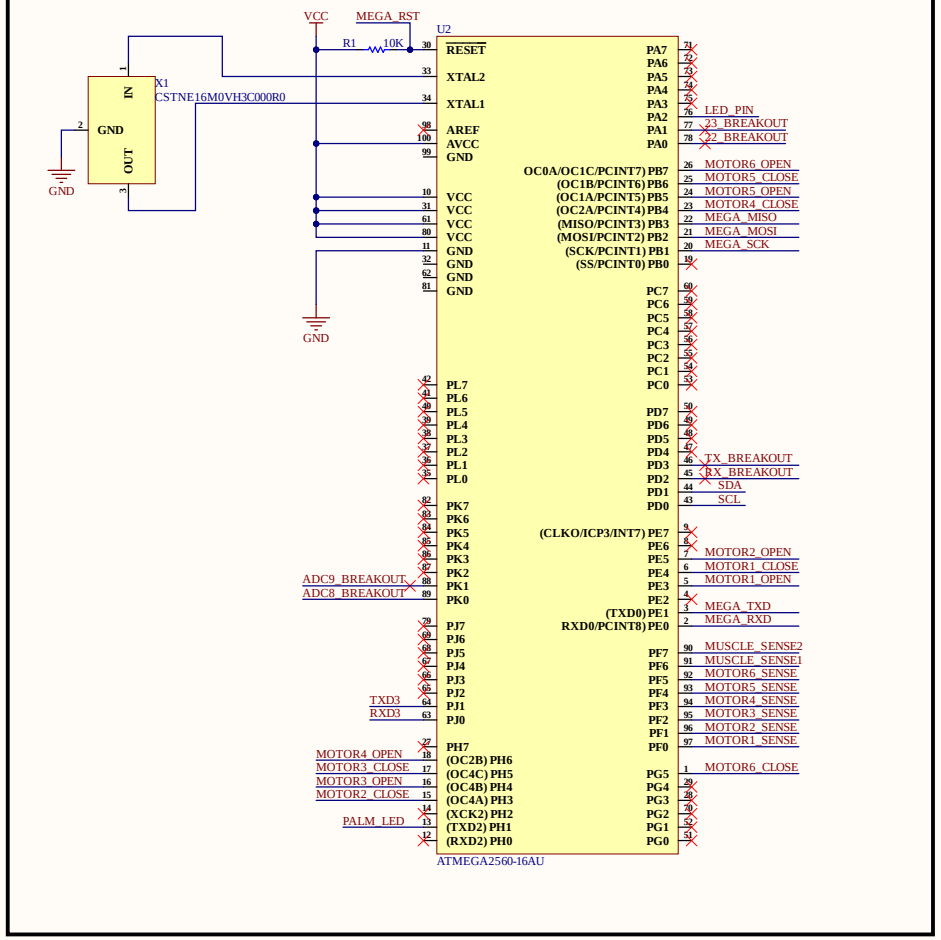
6V TO 5V CONVETER

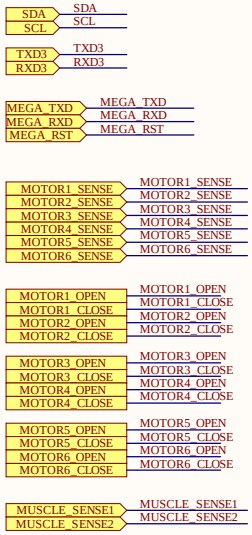


POWER INPUTS

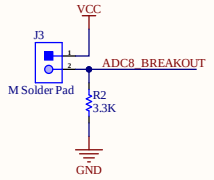


MAIN CONTROLLER

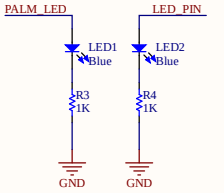




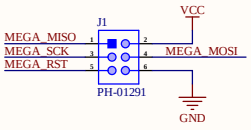
SENSOR CONNECTION

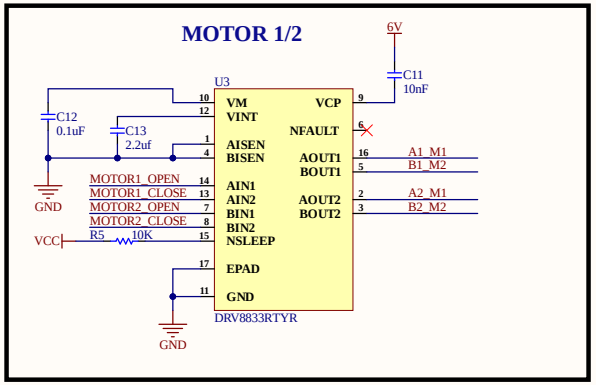


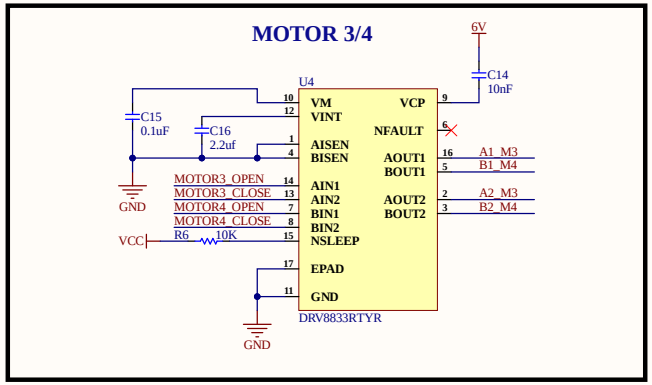
LED'S

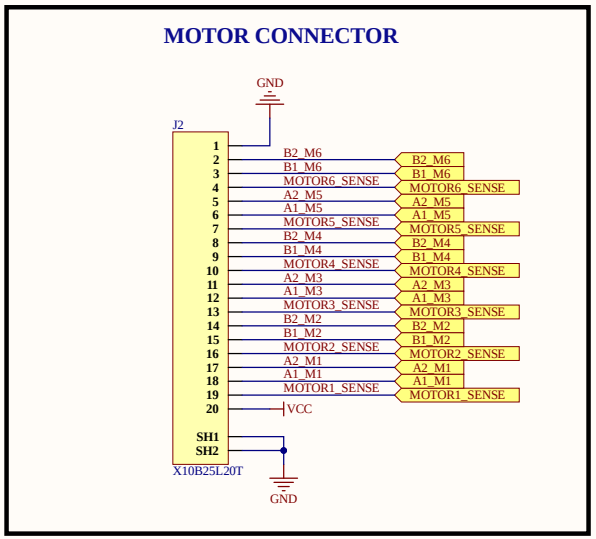


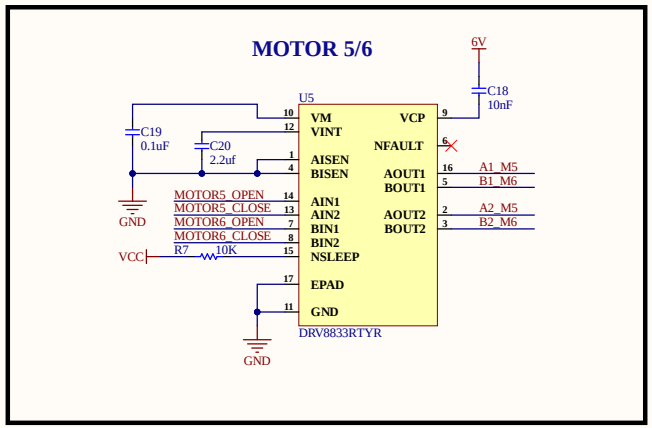
SPI

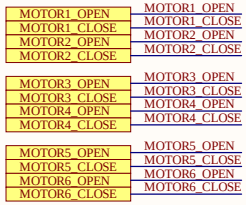


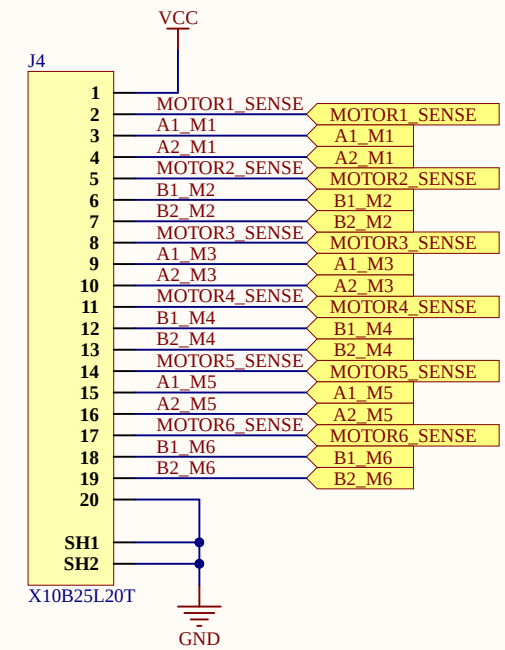

MOTOR 1/2
6V
C11
10nF
U3
C12
0.1uF
C13
2.2uf
VM
VINT
VCP
NFAULT
AISEN
BISEN
AOUT1
BOUT1
A1_M1
B1_M2
MOTOR1_OPEN
MOTOR1_CLOSE
MOTOR2_OPEN
MOTOR2_CLOSE
AIN1
AIN2
BIN1
BIN2
AOUT2
BOUT2
A2_M1
B2_M2
GND
VCC
R5
10K
NSLEEP
EPAD
GND
DRV8833RTYR

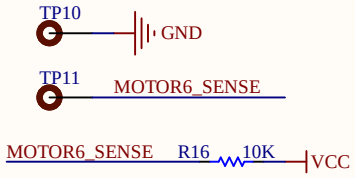

MOTOR 3/4
6V
C14
10nF
U4
C15
0.1uF
C16
2.2uf
VM
VINT
VCP
NFAULT
AISEN
BISEN
AOUT1
BOUT1
A1_M3
B1_M4
MOTOR3_OPEN
MOTOR3_CLOSE
MOTOR4_OPEN
MOTOR4_CLOSE
AIN1
AIN2
BIN1
BIN2
AOUT2
BOUT2
A2_M3
B2_M4
GND
VCC
R6
10K
NSLEEP
EPAD
GND
DRV8833RTYR

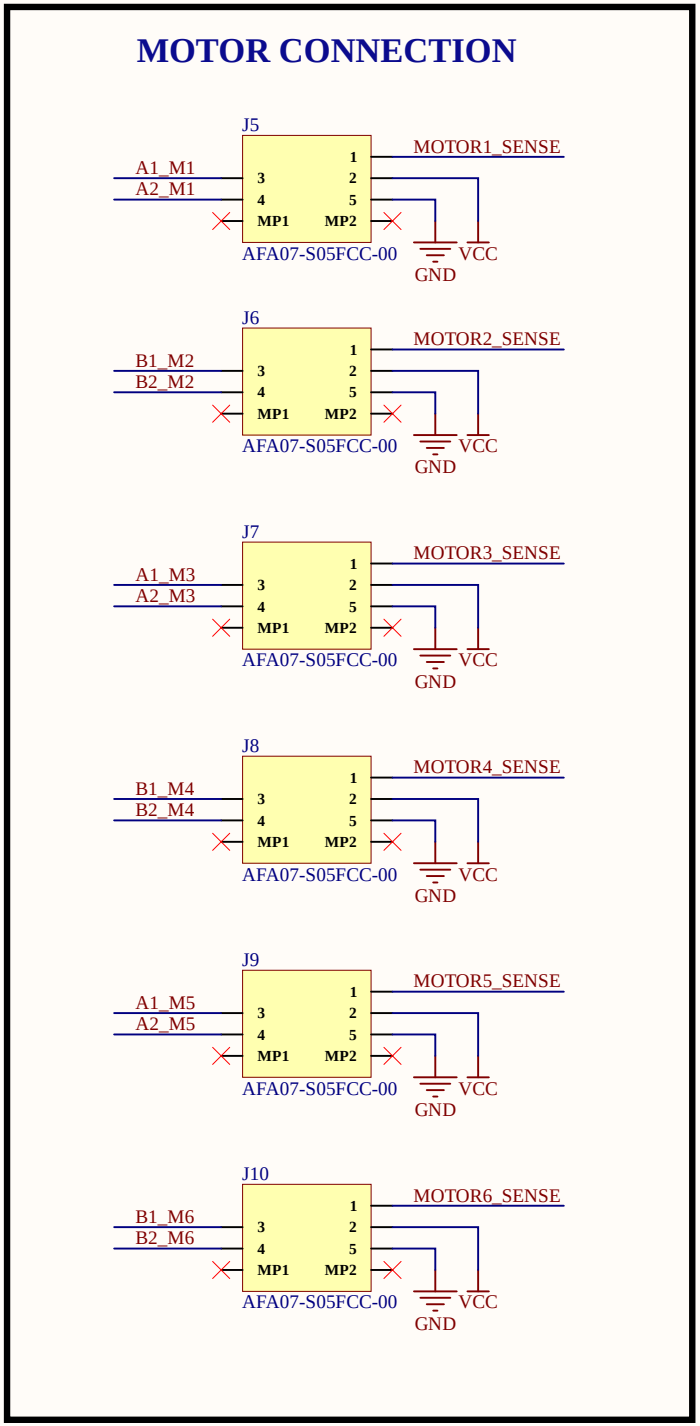

MOTOR CONNECTOR
GND
J2
B2_M6
B1_M6
MOTOR6_SENSE
A2_M5
A1_M5
MOTOR5_SENSE
B2_M4
B1_M4
MOTOR4_SENSE
A2_M3
A1_M3
MOTOR3_SENSE
B2_M2
B1_M2
MOTOR2_SENSE
A2_M1
A1_M1
MOTOR1_SENSE
VCC
SH1
SH2
X10B25L20T
GND


MOTOR 5/6
6V
C18
10nF
U5
C19
0.1uF
C20
2.2uf
VM
VINT
VCP
NFAULT
AISEN
BISEN
AOUT1
BOUT1
A1_M5
B1_M6
MOTOR5_OPEN
MOTOR5_CLOSE
MOTOR6_OPEN
MOTOR6_CLOSE
AIN1
AIN2
BIN1
BIN2
AOUT2
BOUT2
A2_M5
B2_M6
GND
VCC
R7
10K
NSLEEP
EPAD
GND
DRV8833RTYR

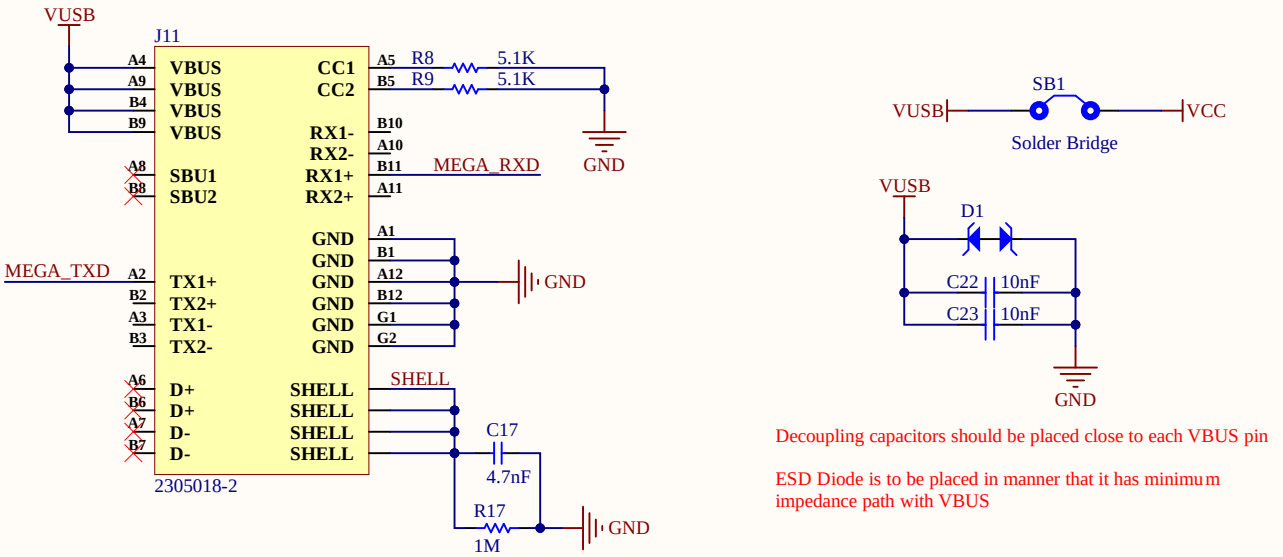

MOTOR1_OPEN
MOTOR1_CLOSE
MOTOR2_OPEN
MOTOR2_CLOSE
MOTOR1_OPEN
MOTOR1_CLOSE
MOTOR2_OPEN
MOTOR2_CLOSE
MOTOR3_OPEN
MOTOR3_CLOSE
MOTOR4_OPEN
MOTOR4_CLOSE
MOTOR3_OPEN
MOTOR3_CLOSE
MOTOR4_OPEN
MOTOR4_CLOSE
MOTOR5_OPEN
MOTOR5_CLOSE
MOTOR6_OPEN
MOTOR6_CLOSE
MOTOR5_OPEN
MOTOR5_CLOSE
MOTOR6_OPEN
MOTOR6_CLOSE


VCC
J4
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
SH1
SH2
X10B25L20T
MOTOR1_SENSE
A1_M1
A2_M1
MOTOR2_SENSE
B1_M2
B2_M2
MOTOR3_SENSE
A1_M3
A2_M3
MOTOR4_SENSE
B1_M4
B2_M4
MOTOR5_SENSE
A1_M5
A2_M5
MOTOR6_SENSE
B1_M6
B2_M6
GND


TP10
GND
TP11
MOTOR6_SENSE
MOTOR6_SENSE
R16
10K
VCC


MOTOR CONNECTION
J5
A1_M1
A2_M1
MOTOR1_SENSE
AFA07-S05FCC-00
VCC
GND
J6
B1_M2
B2_M2
MOTOR2_SENSE
J7
A1_M3
A2_M3
MOTOR3_SENSE
J8
B1_M4
B2_M4
MOTOR4_SENSE
J9
A1_M5
A2_M5
MOTOR5_SENSE
J10
B1_M6
B2_M6
MOTOR6_SENSE

USB-C CONNECTOR



AUDIO CONNECTION

